GREEN SPINE

AEC NL HACKATHON 2026

A NATIVE CORRIDOR FROM
FELLENOORD TO THE DOMMEL
EINDHOVEN · 13.11.2026

THE TEAM



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Architect and computational designer

Corridor design, placement model, visuals



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Structural engineering

Soil volumes, no-plant zones, KPI formulas



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Structural engineering

Tree classes, buildability, Phasing, canopy and pergola



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AI and data

Pipeline, scoring, species ranking, dashboard

Fellenoord is being rebuilt now. Green corridors succeed above ground and fail below it, and their results are almost never measured.

01

NOW

Fellenoord is paved, road-dominated and about to be redeveloped. The window to decide where the green goes is open today

02

BELOW GROUND

Street trees die from too little rootable soil, compaction and salt, not from bad design intent

03

UNMEASURED

Street greening projects state many goals; few are ever evaluated against them

Ontwikkelvisie Fellenoord; Toronto soil cell study (The Nature of Cities); Re-imagining Streets, University of Melbourne.

WHO IT IS FOR

THE CITY PLANNER	Needs to defend where the planting budget goes, against targets, with evidence	**A ranked, costed planting schedule**
THE RESIDENT	Walks from the station to the Dommel and wants shade, not asphalt	**A continuous shaded route**
THE DEVELOPER	Builds cluster by cluster to 2040 and must deliver green per plot	**A phased plan that fits the cluster passports**

One tool, three readings of the same result.

STAKEHOLDERS

WHO	ROLE AT FELLENOORD	WHAT THEY WANT FROM THE SPINE
Gemeente Eindhoven	Owns the public space and the climate policy	Evidence that greening targets are met
Residents	Occupy	
ProRail and NS	Station and bus interchange	Comfort for 180,000 daily travellers by 2040
Real Estate Developers	Build the clusters, about 1,000 homes on one site	Clear, costed green obligations per plot
Residents, now and 2040		

Ontwikkelvisie Fellenoord; Provincie Noord-Brabant; ProRail (2023); BAM press release (July 2026).

REFERENCES WE STUDIED

PROJECT	WHAT IT IS	MEASURED RESULT
Medellín, Corredores Verdes	30 green corridors, 36 km, from 2016	About 2 °C cooling, per city data
Barcelona, Green Axis	Seven street segments and four squares redesigned	Before and after study: better greenery, safety, walking
Toronto, Bloor Street	Engineered soil cells under street trees	Tree deaths from several cumulative causes
Milwaukee street trees	Survival tracked from 1979 to 2005	Narrower planting strips meant more deaths
Eindhoven, Vestdijk	Car street turned into a green boulevard	Improved soil and infiltration, less heat stress

BBC Future Planet via Jagonews; Pérez et al. (Eur. J. Public Health); The Nature of Cities; Urban Forestry & Urban Greening; UNaLab.

LESSONS LEARNT: WHAT TO AVOID

THE FAILURE	WHERE IT HAPPENED	WHAT WE DO ABOUT IT
Trees in too little soil	Toronto and Milwaukee: pits of a few square metres, roots lifting paving, higher mortality in narrow strips	Every tree checked against a soil volume rule for its size class. If the ground cannot carry it, the tool says so
Maintenance underfunded	Medellín: stretches left overgrown, upkeep about 625,000 dollars a year	Native, low-maintenance species; the schedule reports how many trees need engineered soil
Benefits never measured	A review of five international retrofits found canopy, biodiversity and social value rarely evaluated	Four KPIs computed at 2026, 2040 and 2060, exported with the plan
Greening pushes people out	Barcelona: researchers flag likely gentrification after the Green Axis	We prioritise the social-rent clusters and report who the shade reaches
A pretty image, not a plan	Common to hackathon entries and to visions	Output is a planting schedule with positions, species and phasing

The Nature of Cities; Milwaukee street tree study; Rio Negro / BBC on Medellín; Re-imagining Streets (Melbourne); Camerin & Longato (2025).

KPIs AND METRICS

KPI	HOW WE CALCULATE IT	TARGET
Canopy cover	Merged crown area divided by corridor area	30% (3-30-300 rule)
Shaded route	Walking centreline under canopy, divided by total length	60% (our proposal)
Longest gap	Longest uncovered stretch of the centreline, in metres	25 m or less (our proposal)
Trees in natural soil	Trees whose cell meets their soil need, divided by all trees	60% (our proposal)
Species diversity	Share of the largest species, genus and family	10-20-30 rule

Also reported: tree count, species count and engineered soil volume in m^3 as a cost proxy.

3-30-300 rule (Konijnendijk); 10-20-30 rule (Santamour). Shade, gap and soil targets are our own proposals.

THE STUDY CASE: FELLENOORD

249,783
people in Eindhoven, 1 January 2026

~1,000
homes on one acquired site, 85% affordable

6,000–7,000
homes planned on Fellenoord by 2040

30%
target share below the social rent threshold

A few hundred people live in the area today. By 2040 it becomes a dense city-centre district next to a station carrying 180,000 travellers a day. Eindhoven as a whole counts 40,676 residents under 18 and 40,771 over 65, the two groups most exposed to summer heat.

CBS via business.gov.nl (2026); citypopulation.de age groups (2025); Eindhoven News; BAM (2026); ProRail (2023).

DATA COLLECTION

Add an example of a plant entry, list sources

RAW DATA AND HOW WE COLLECT IT

LAYER	SOURCE	STATUS
Plant database	Team's own merged species table	In the shared folder
Buildings in 3D	3D BAG, four CityJSON tiles	In the shared folder
Policy and KPIs per plot	Ontwikkelvisie and cluster passports 1A, 1B, 2, 9B, 9C	In the shared folder
Existing trees	Eindhoven open data, Bomen, public domain	Download, 10 minutes
Public green	Eindhoven open data, Openbaar groen	Download, 10 minutes
Surfaces and roads	OpenStreetMap	Download, 15 minutes
Soil and heat	No open layer at street resolution	Proxy, stated as an assumption

data.eindhoven.nl; 3dbag.nl; OpenStreetMap contributors (ODbL); Klimaateffectatlas (download options limited).

SOIL

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

SOIL

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX



HOW WE USED DATA

HOW WE USE THE DATA

1	**CLIP AND ALIGN**	Every layer cut to the corridor and converted to the Dutch national grid, EPSG:28992, so metres are metres
2	**GRID**	The corridor's public space divided into 2 m cells
3	**ATTRIBUTE**	Each cell inherits blocked, paved, soil, heat and distance to existing canopy
4	**ONE FILE**	All of it becomes a single scored grid, the only handover between the data work and the design model

Utilities, traffic and pedestrian flow are stated assumptions, not hidden guesses.

ALGORITHMIC WORKFLOW

INPUTS

3D mass, surfaces, greenery, soil, climate, plants

→

SCORE

Each cell: heat + soil + gap to green

→

PLACE

Best cell first, largest tree the soil allows, spacing by crown

→

SPECIES

Filter by soil and size, enforce 10-20-30

→

MEASURE

Four KPIs at 2026, 2040, 2060

LOOP: rerun with three weight sets — most shade, most connected, lowest cost — to generate options A, B and C, then rank by targets met

A transparent scoring rule, not a black box. Every tree can be traced back to the cell score that put it there.

WHAT WE ARE BUILDING

“Based on the Fellenoord urban layout, generate vegetation planting plans optimizing the ecosystem and infrastructure impact, then show canopy cover and habitat connection today, in 2040 and in 2060.”

GENERATES	Three planting strategies on the same corridor, from one scoring rule	**RESPECTS**	Soil, utilities and the carriageway. Trees only where the ground can carry them
MEASURES	Four KPIs at 2026, 2040 and 2060, against targets	**DELIVERS**	A planting schedule the city can act on, not only an image

Not a rendering. A tool the jury can move a slider on.